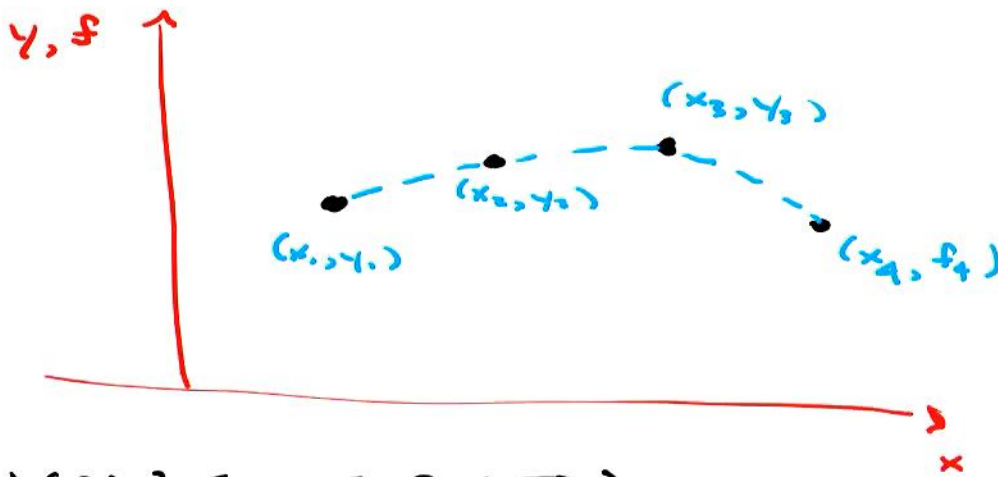


POLYNOMIAL INTERPOLATION



1. LAGRANGE INTERPOLATION

- CONSIDER STANDARD PROBLEM FROM NUMERICAL ANALYSIS

$$\text{NOTATION } (x_i, y_i) \equiv (x_i, f(x_i)) \equiv (x_i, f_i)$$

- GOAL: GIVEN n DATA PTS

$$(x_j, f_j), j = 1, 2, \dots, n$$

FIND (CONSTRUCT) UNIQUE POLYNOMIAL OF MAXIMUM DEGREE $n-1$ PASSING THROUGH ALL DATA PTS

(DEGREE $\equiv$ LARGEST POWER OF INDEPENDENT VAR.)

$$p(x) = \sum_{i=0}^{n-1} c_i x^i$$

(POLYNOMIAL OF DEGREE $n-1$)

- ONCE $p(x)$ IS DETERMINED, CAN USE TO GET APPROXIMATE (INTERPOLATED) VALUES OF $f(x)$ FOR $x_1 \leq x \leq x_n$
- MANY REPRESENTATIONS OF INTERPOLATING POLY. LAGRANGE APPROACH IS TO TAKE

$$p(x) = \sum_{j=1}^n f_j \underline{l_j(x)}$$

($\underline{l_j(x)}$ POLY. OF DEGREE $n-1$)

WHERE $\underline{l_j(x)}$ ARE CHARACTERISTIC POLYNOMIALS SATISFYING

$$l_j(x_i) = \delta_{ji}$$

δ_{ji} IS KRONECKER DELTA FCN

$$\delta_{ji} = \begin{cases} 1 & j=i \\ 0 & j \neq i \end{cases}$$

THEN

$$p(x_i) = \sum_{j=1}^n f_j l_j(x_i) = \sum_{j=1}^n f_j \delta_{ji} = f_i$$

AS DESIRED

- NEED TO BUILD CHARACTERISTIC POLYNOMIALS.
FOLLOWING DOES THE TRICK

$$l_j(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_{j-1})(x-x_{j+1})\dots(x-x_n)}{(x_j-x_1)(x_j-x_2)\dots(x_j-x_{j-1})(x_j-x_{j+1})\dots(x_j-x_n)}$$

IF $x = x_j$ THEN NUM = DEN SO $l_j(x_j) = 1$

IF $x = x_i$, $i \neq j$ THEN EXACTLY ONE TERM IN NUM IS 0 SO $l_j(x_i) = 0$

- FINAL FORMULA FOR LAGRANGE POLY.

$$p(x) = \sum_{j=1}^n f_j l_j(x) \quad (1)$$

$$l_j(x) = \prod_{i=1, i \neq j}^n \frac{x-x_i}{x_j-x_i} \quad (2)$$

2 BARYCENTRIC INTERPOLATION

- SO CALLED BECAUSE CAN RELATE TO PROBLEM OF DETERMINING CENTRE OF MASS FOR COLLECTION OF PARTICLES WITH GIVEN WEIGHTS, POSITIONS
- FIRST, DEFINE

$$l(x) = (x-x_1)(x-x_2)\dots(x-x_j)\dots(x-x_n)$$

NUMERATOR IN (2) IS $l(x)$ DIVIDED BY $(x-x_j)$

- NEXT, DEFINE BARYCENTRIC WEIGHTS W_j

$$W_j = \frac{1}{\sum_{k=1, k \neq j}^n (x_j - x_k)} \quad (\text{DENOMINATOR OF (2)})$$

THEN CHARACTERISTIC POLYNOMIAL l_j IS

$$l_j(x) = l(x) \frac{W_j}{x - x_j}$$

SO WE GET FIRST FORMULA FOR BARYCENTRIC INTERPOLATION

$$p(x) = l(x) \sum_{j=1}^n \frac{W_j}{x - x_j} f_j \quad (3)$$

• ADVANTAGE

- ORIGINAL $O(n^2)$ CALCS. TO INTERPOLATE TO ANY PT
- BARYCENTRIC: $O(n^2)$ CALCS. TO COMPUTE WEIGHTS;
 $O(n)$ CALCS. TO INTERPOLATE TO ANY PT.

- GET ANOTHER VERSION OF BARYCENTRIC INTERPOLATION BY LETTING THE DATA VALUES f_j BE THE CONSTANT FUNCTION 1. THEN FROM (1) AND (3) WE HAVE

$$1 = \sum_{j=1}^n l_j(x) = l(x) \sum_{j=1}^n \frac{W_j}{x - x_j}$$

- DIVIDE (3) BY LAST RESULT, CANCEL $l(x)$
GET 2ND BARYCENTRIC FORMULA

$$p(x) = \frac{\sum_{j=1}^n \frac{W_j}{x - x_j} f_j}{\sum_{j=1}^n \frac{W_j}{x - x_j}} \quad (4)$$

(2ND FORMULA)

• 3.1 NUMERICAL EXAMPLE

EXERCISE: CONSTRUCT THE POLYNOMIAL PASSING THROUGH

$$(-1, 6), (1, 9), (3, 34)$$

• 3.2 SYMBOLIC EXAMPLE

FIT LAGRANGE POLY TO

$$(-h, f_{-1}), (0, f_0), (h, f_1)$$

AND EVALUATE DERIVATIVE AT $x=0$

$$p(x) = \sum_{j=1}^3 f_j l_j(x)$$

$$= f_{-1} \frac{x(x-h)}{(-h)(-2h)} + f_0 \frac{(x+h)(x-h)}{(h)(-h)} + f_1 \frac{(x+h)(x)}{(2h)(h)}$$

$$= f_{-1} \frac{x^2 - hx}{2h^2} - f_0 \frac{x^2 - h^2}{h^2} + f_1 \frac{x^2 + hx}{2h^2}$$

• DERIVATIVE AT $x=0 \Rightarrow$ PULL OFF TERMS LINEAR IN x

$$\left. \frac{dp}{dx} \right|_{x=0} = \frac{f_1 - f_{-1}}{2h}$$

WE'LL SEE THIS FORMULA AGAIN WHEN DISCUSSING
FINITE DIFFERENCE METHODS